Complex Data Analysis - MathOverflow

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Abstract

This article explores a dataset of interactions in a Q&A forum, MathOverflow, and presents several statistical analyses of the data.

Keywords: Network Analysis, MathOverflow, SNAP Dataset, Social Networks, Graph Mining, Community Detection

1 Introduction

Social networks in the real world contain a lot of information encoded within their interactions. By studying this data, it will be possible to better understand several questions, such as how communities form or how information disseminates through a network. This is the motivation behind the analysis of this dataset.

MathOverflow is a suitable case study because, being a Q&A forum where users discuss and answer each other about math questions, it leads to the creation of several communities and several other interesting patterns related to information spread.

The goals of this assignment are to create a representative subset of the network, compute key structural metrics (such as reciprocity and centrality), compare the results with existing literature, and visualize the community structure.

Project Repository. All source code, data subsets, notebooks, and Gephi visualizations used in this report are publicly available at:
https://github.com/jotanmiguel/ProjetoADC-Grupo12.

2 Related Work

2.1 MathOverflow as a Scientific and Social Network

MathOverflow, MO, is an online Q&A platform targeted at professional mathematicians, graduate students and researchers. Unlike general-purpose Q&A communities, its purpose is explicitly academic: users post advanced research-level questions, discuss partial results, exchange counterexamples and provide references to the mathematical literature. More than 90% of frequent contributors, most of them males with an average age of 32, have formal mathematical training, and many use their real names, treating the platform as an extension of their professional identity [3]. The presence of established mathematicians, shapes the culture of the site: contributions are expected to be rigorous, concise and supported by citations to the literature.

MO can be modeled as a directed interaction network in which nodes represent users and edges denote knowledge transfer events such as answers, clarifications or supporting mathematical observations. This structure encodes the flow of expertise within the community and reveals an informal hierarchy shaped by domain knowledge and academic seniority. As shown by Montoya et al.[1], structural centrality measures such as degree and betweenness are strongly correlated with indicators of social achievement on the platform, including reputation, upvotes and visibility. Users who occupy central positions in the network tend to be highly knowledgeable contributors whose answers attract further engagement and validation.

The collaborative nature of the platform has also been emphasized by empirical studies: solutions often emerge from the incremental contributions of multiple participants, each providing partial insights that jointly resolve a research question [2]. This behavior illustrates how MO functions as a distributed system of collective mathematical intelligence.

Overall, MathOverflow occupies a unique position between a social network and a scientific collaboration platform. Its interaction patterns are shaped by disciplinary expertise, reputation mechanisms and the conventions of mathematical communication. These characteristics make MO an ideal case study for network analysis, providing a rich environment in which to examine hierarchical structure, patterns of collaboration and the dynamics of knowledge exchange.

2.2 Structural Properties of the MathOverflow Network

The structural behavior of MathOverflow has been studied extensively through the lens of network analysis. One of the recurrent findings is that the platform exhibits a heavy-tailed degree distribution: most users contribute infrequently, while a small number of highly active experts are responsible for a disproportionate fraction of all answers [3]. This distribution reflects the underlying inequality in mathematical expertise and creates a naturally hierarchical structure within the network.

The network is directed and strongly asymmetric. High-degree users typically answer questions posed by low-degree users, a pattern characteristic of disassortative networks. This hierarchical organization aligns with the results of [**?**], who show that structural centrality measures such as degree and betweenness correlate strongly with indicators of social achievement on the platform, including reputation, visibility and community recognition. Experts occupy central positions in the graph and their answers tend to attract further engagement from other participants.

MO also displays modular and locally cohesive structure. The clusters often have the same structure as the original network [?]. Interactions tend to cluster by mathematical subfield, forming communities that correspond to topical areas. As shown by Tausczik et al.[2], collaborative reasoning often emerges within these clusters: solutions are built incrementally, with multiple users contributing partial insights or complementary arguments. These patterns mirror the collaborative practices of mathematical research and further reinforce the interpretation of MO as a scientific collaboration network rather than a casual social platform such as Reddit[1] or Quora[2].

Overall, the literature portrays MathOverflow as a sparse, hierarchical and highly modular directed network, dominated by a small set of expert contributors and characterized by asymmetric knowledge flow, distributed problem solving and topically organized communities.

2.3 Expected Behavior According to the Literature

Prior studies of MathOverflow report several structural patterns that are characteristic of expert-driven knowledge networks. First, user activity follows a heavy-tailed distribution, with a small minority of highly active experts producing a disproportionately large share of all answers [3]. This results in a hierarchical structure with clear hubs.

Second, MathOverflow is known to be disassortative: high-degree users tend to interact with low-degree users, reflecting an expert-to-novice knowledge flow [?]. Low reciprocity is also expected, as interactions are primarily unidirectional (A answering B), rather than conversational.

Third, the network is sparse but modular. Communities tend to form around mathematical subfields, and collaborative reasoning frequently occurs within these topical clusters [2]. Clustering is therefore higher than in a random directed graph, but still relatively low compared to social friendship networks.

These properties—heavy-tailed degree distributions, negative assortativity, low reciprocity, and moderate modularity—serve as a theoretical baseline for evaluating whether our filtered subset remains representative of the structural behaviour documented in the literature.

3 Dataset Description

From the Stanford Network Analysis Project (SNAP)[3], we've chosen `sx-mathoverflow` as our dataset. MathOverflow[4] is a question-and-answer (Q&A) platform targeted at professional mathematicians and researchers, where users post research-level problems and exchange knowledge. The SNAP dataset captures interactions between users extracted from the public Stack Exchange data dumps [1].

3.1 Structure of the SX-MathOverflow Dataset

The full dataset represents MathOverflow as a directed temporal network. Each node corresponds to a user account, and each temporal edge is a triplet (u, v, t) indicating that user u interacted with user v at timestamp t. Interactions include:

- answering another user's question,
- commenting on a question,
- commenting on an answer.

The complete temporal dataset contains **24,818 nodes** and **506,550 time-stamped interactions** in a period of 2,350 days. A static projection of all interactions produces a directed graph with 239,978 edges.

SNAP additionally provides three derived interaction layers, each isolating a specific type of user activity:

- `sx-mathoverflow-a2q`: answer-to-question interactions,
- `sx-mathoverflow-c2q`: comment-to-question interactions,
- `sx-mathoverflow-c2a`: comment-to-answer interactions.

4 Subset Creation and Justification

The original Math Overflow interaction dataset ($N_{\text{nodes}} = 24,818$; $N_{\text{temporal edges}} = 506,550$) is extensive. To focus the analysis on the most representative dynamics of the community and ensure computational tractability for complex network metrics, a two-step procedure was implemented to create a representative subset.

The adopted methodology combines: (1) Temporal Filtering, (2) User Activity Filtering.

4.1 Temporal Filtering

The first step involved isolating a specific time window that captures a phase of high activity and network consolidation within the platform. The analysis of the temporal evolution of the number of interactions (temporal edges) over time is presented in Figure 1.

The graph clearly shows a significant spike in interaction volume between 2010 and 2011, followed by a period of relative stabilization. Based on this observation, the initial subset was delimited to the period from January 1, 2010, to December 31, 2012 (we decided to extend the second date to get more data).

This initial action reduced the dataset to 12,470 nodes (50% reduction) and 266,094 temporal edges (48% reduction).

[1] https://www.reddit.com/
[2] https://pt.quora.com/
[3] https://snap.stanford.edu/data/sx-mathoverflow.html
[4] http://MathOverflow.net

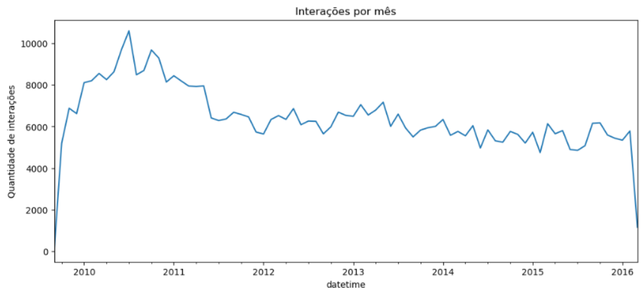

Figure 1. Temporal evolution of the volume of interactions (edges) in the Math Overflow dataset. The graph clearly shows a spike in growth between 2010 and 2011.

4.2 User Activity Filtering

The second step applied an activity filter to isolate the core participants of the community, thereby excluding inactive or "one-shot" users whose peripheral activity could distort structural metrics.

- **Procedure:** The total degree (sum of in-degree and out-degree) was calculated for all users within the temporal subset.
- **Criterion:** Only users with a total degree greater than 25 interactions were retained.
- **Justification:** This minimum threshold defines a core set of active contributors. It ensures that the resulting induced subgraph reflects repeated and meaningful interactions, where knowledge transfer and expertise are most likely to be exhibited.

The final edge list retained only interactions where both the source (*src*) and the target (*dst*) nodes satisfied this minimum activity criterion.

4.2.1 Filtered Graph Statistics.
After applying the activity filter, the graph was constructed, containing:

- **Nodes (Users): 2457**
- **Edges (Static Interactions): 88294**

The static undirected version of this graph is fully connected, which means that its largest weakly connected component represents the entirety of the graph.

We can see a visualization of our subset via gephi in figure 2.

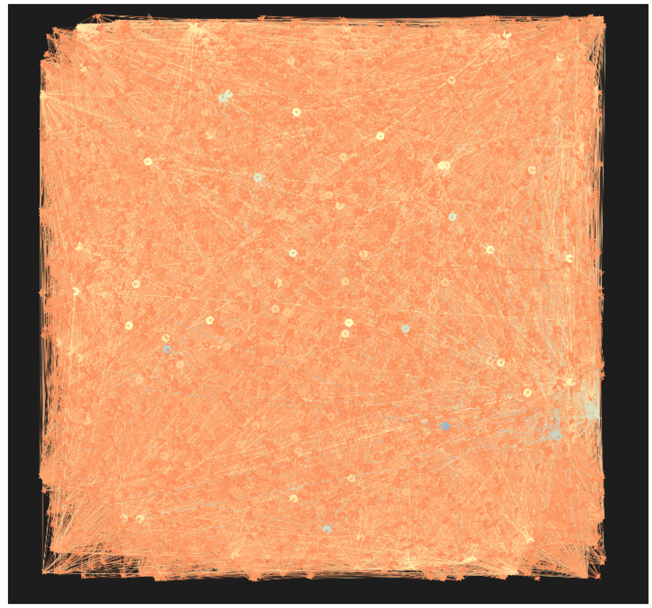

Figure 2. Visualization of the chosen subset.

5 Subset Analysis: Structural Properties

This section presents the structural analysis of the MathOverflow network subset (Largest Weakly Connected Component - LCC), focusing on the key properties of density, clustering coefficient, and connectivity.

5.1 Fundamental Statistics and Network Sparsity

The fundamental statistics quantify the basic structure of the active community:

- **Clustering Coefficient (C): 0.0970**
- **Density (ρ): 0.0146**

Density Interpretation: The network's density indicates that the network is sparse (i.e., not fully connected relative to its potential size). This characteristic is common in real-world knowledge networks, where interactions are specific and localized, rather than universal.

5.2 Clustering Coefficient, Reciprocity, and Flow Direction

Clustering Coefficient. The network exhibits a Clustering Coefficient of **0.0970**. This non-random value signifies a higher-than-expected tendency for the formation of closed triangles within the active subset. This high local clustering indicates the formation of local communities where users often interact around shared topics or experts, demonstrating high proximity among interconnected members.

Average Path Length (APL) and Connectivity. The Average Shortest Path Length (APL) could not be computed for the directed graph due to its strong asymmetry. However, when considering the undirected projection of the graph, the APL is **2.25**.

> *This result reveals a crucial network property: the active community is a Small-World network, where any user can reach any other user through an average of only* 2.25 *steps (ignoring direction). This demonstrates high global efficiency in information access.*

Despite this efficiency, the directed APL calculation failed because the subset, while being the Largest Weakly Connected Component (LWCC), is not a Strongly Connected Component (SCC). The existence of a path A $\rightarrow$ B (User A answers B's question) does not guarantee the existence of a path B $\rightarrow$ A. This directional constraint prevents the calculation of a global shortest path, reinforcing the one-way nature of knowledge transfer inherent in the Q&A system.

Reciprocity. The Reciprocity of the network is **0.3883**. Reciprocity measures the probability that a directed edge is part of a reciprocal pair (A $\rightleftharpoons$ B).

- This value confirms that the majority of relationships ($\sim$ 61%) remain unidirectional, supporting the initial hypothesis of a "knowledge provider $\rightarrow$ seeker" or "Expert $\rightarrow$ Novice" dynamic.
- However, the **38.83%** reciprocity also indicates that a substantial portion of the relations among the core contributors are mutual. This suggests that high-activity users often exchange roles, asking research-level questions as well as providing answers, leading to a significant mutual exchange of knowledge within the active community.

Out-In Degree Assortativity ($r_{out\text{-}in}$). The Out-In Degree Assortativity coefficient is $\mathbf{r}_{\text{out-in}} = \mathbf{-0.1161}$. This coefficient measures the correlation between the Out-Degree (k_{out}) of the source node (the answer provider) and the In-Degree (k_{in}) of the target node (the question poster).

- The negative value ($\mathbf{-0.1161}$) signifies disassortativity. This confirms the network's Scale-Free structure and hierarchical organization, as high-degree nodes tend to connect to low-degree nodes.
- **Knowledge Dispersal:** Specifically, the negative correlation indicates that the most prolific knowledge providers (high k_{out} authorities) preferentially connect to users who are the biggest knowledge solicitators (high k_{in} demand) and peripheral users alike.

Degree Assortativity (r). The analysis of assortativity for the different degree combinations reveals a dominant characteristic of **disassortativity** (negative correlation) across all combinations:

- **Out-In Assortativity ($\mathbf{r}_{\text{out-in}}$): $\mathbf{-0.1161}$**
- **Out-Out Assortativity ($\mathbf{r}_{\text{out-out}}$): $\mathbf{-0.1106}$**
- **In-In Assortativity ($\mathbf{r}_{\text{in-in}}$): $\mathbf{-0.0896}$**

The negative value of $\mathbf{r}_{\text{out-in}}$ had already established the network's hierarchy: the greatest knowledge providers (high k_{out}) link to the greatest solicitors (high k_{in}) and to peripheral users.

> *Even more revealing is the fact that $\mathbf{r}_{out\text{-}out}$ and $\mathbf{r}_{in\text{-}in}$ are also negative. This indicates that even interactions between peers (whether between answerers or between question-posters) follow a disassortative pattern. In social networks, it is typically expected for hubs to connect to hubs (assortativity). In MathOverflow, this generalized disassortativity confirms that the dominant dynamic is the **dispersion of knowledge** rather than collaboration between peers of equal status.*

Specifically:

- Out-Out Assortativity ($\mathbf{-0.1106}$): The most prolific answerers (high k_{out}) do **not** tend to answer questions posed by other prolific answerers. This suggests that interactions are more about the **transfer of knowledge to the base** than about high-level technical debates among experts.
- In-In Assortativity ($\mathbf{-0.0896}$): The biggest information solicitors (high k_{in}) do **not** tend to ask questions of other solicitors. This metric has a less extreme value but points to a structure where questions are directed by thematic necessity, and not by an affinity of "seeker" status.

Central Conclusion: The combination of low reciprocity and generalized disassortativity is the clearest evidence that the primary function of the MathOverflow subset is the efficient dispersion of knowledge from centers of authority to the user base.

5.3 Degree Analysis

Average and Maximum Degree: The degree analysis reveals significant heterogeneity in user activity:

- **Average In-Degree ($\langle k_{in} \rangle$): 35.94**
- **Average Out-Degree ($\langle k_{out} \rangle$): 35.94**
- **Maximum In-Degree ($\max k_{in}$): 498** (The most popular question seeker).
- **Maximum Out-Degree ($\max k_{out}$): 792** (The most prolific knowledge provider).

The large disparity between the average degree ($\sim$ 36) and the maximum degrees (up to 792), coupled with $\max k_{out} > \max k_{in}$, strongly suggests that the network is "Scale-Free" and that knowledge supply is more concentrated among a few authorities than knowledge demand.

The graphs represented in the figures 3 and 4 help visualize just how much the degrees of the users spread.

5.4 Centrality Analysis

To identify the most structurally critical users in mediating knowledge flow, we computed the Betweenness.

Betweenness. Betweenness measures the extent to which a user lies on the shortest paths between all other pairs of users. High scores identify knowledge brokers who are essential for connecting disparate parts of the network.

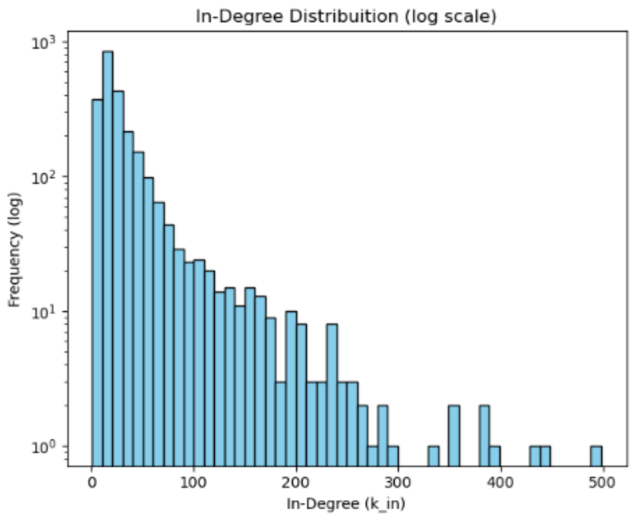

Figure 3. In Degree Distribution.

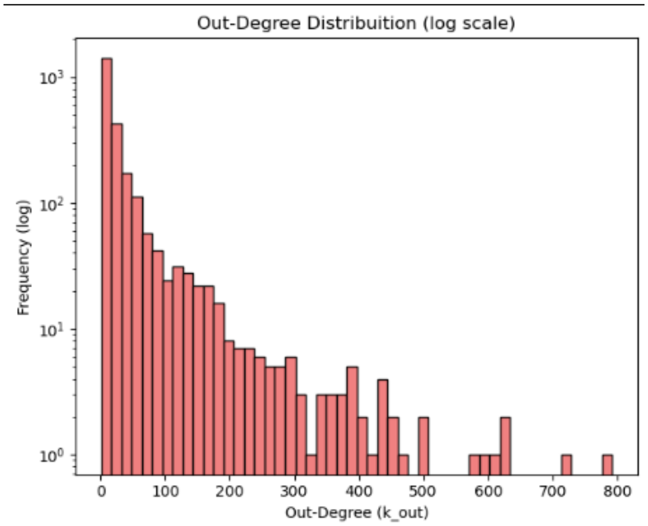

Figure 4. Out Degree Distribution.

The analysis revealed a strong concentration of "betweenness power" in a small set of users. The top five users by Betweenness Centrality are presented in Table 1.

Table 1. Top 5 Users by Betweenness Centrality.

User_ID	Betweenness Centrality
290	0.0515
11142	0.0480
297	0.0353
1409	0.0343
6094	0.0310

The user with the highest score, User ID 290 (**0.0515**), mediates over 5% of all shortest paths in the network. These results highlight a pattern where structural importance is concentrated among a few strategic nodes.

Crucially, users with high Betweenness Centrality often act as structural bridges that link the different communities, ensuring that the specialized knowledge produced in one subgroup can be efficiently transmitted across the network. This brokerage role is distinct from the high volume contribution quantified by the maximum Out-Degree (max k_{out}).

5.5 Community Detection

To identify the underlying structure and grouping within the core user community, the Louvain Algorithm was applied to the static subset graph. The results of the community detection are summarized below:

- **Global Modularity (Q):** 0.2451
- **Total Number of Communities:** 7
- **Size of the Largest Community:** 549 nodes
- **Size of the Second Largest Community:** 509 nodes

The resulting Modularity value of ($Q = 0.2451$) suggests a moderate but observable community structure. While the value is below the frequently cited threshold of 0.3 or 0.4 for very strong structures, it is significantly greater than zero, confirming that users tend to interact more frequently within certain groups than across the entire network.

The algorithm identified 7 distinct communities. The network structure is characterized by its two largest communities, which are composed by 549 and 509 nodes, respectively, suggesting that the active user base is predominantly separated into two main, dense groups, with the remaining nodes distributed among five smaller groups. This heavy concentration in two main components may explain the moderate modularity, as these large communities likely share a substantial number of inter-community edges.

We can visualize the communities present in Figure 5.

5.6 Temporal Activity Patterns

Since our dataset has a temporal feature we decided to take advantage of that and explore it furthermore. The graph in figure 6 was made to better understand the temporal activity patterns.

The data given by the graph seems non-intuitive since it reveals a highly distinctive pattern: interaction volume peaks between 00:00 and 04:00 UTC and reaches its minimum between 13:00 and 16:00 UTC. This inverted cycle is atypical for a professional, research-oriented platform such as MathOverflow. A plausible explanation is that a substantial portion of the platform's most active users are located in Asian time zones. When converted, the global peak at around 01:00–02:00 UTC corresponds to the beginning of the workday in India (06:00–07:00) and mid-morning in China and Singapore (09:00–10:00). This aligns well with the demographic composition of the mathematical research community in

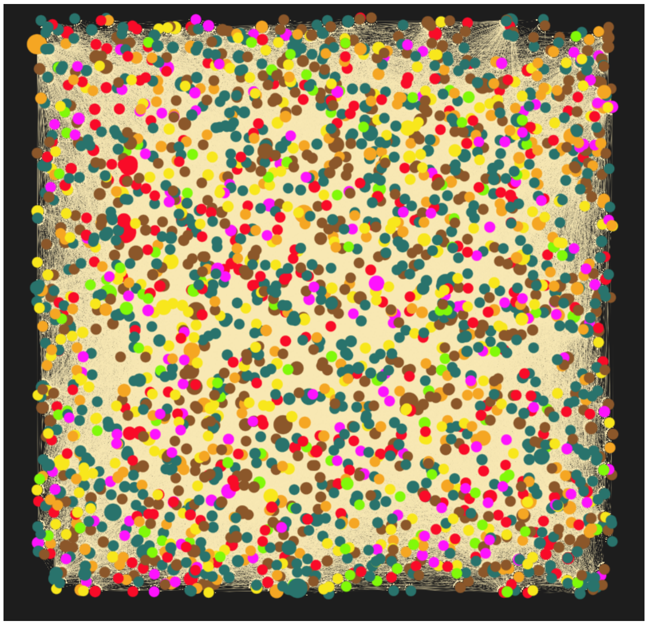

Figure 5. Communities.

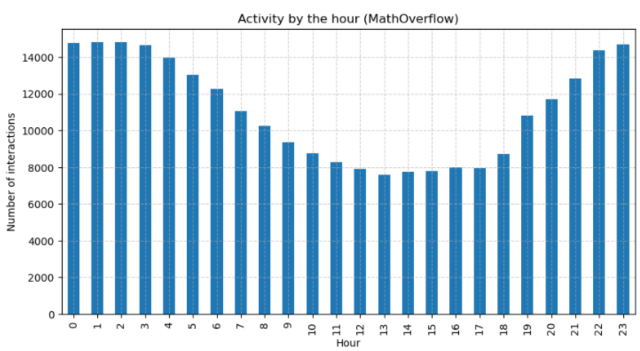

Figure 6. Hourly interaction volume (UTC).

these regions. Another plausible and much simpler explanation is that the high activity during those hours can be attributed to users in the USA, where MathOverflow has a large user base, who have more time to dedicate to the site during the night.

6 Discussion

6.1 Comparison With Reported MathOverflow Properties

Prior studies describe the platform as a sparse, expert–driven knowledge network characterized by uneven participation, hierarchical structure and topic–based modularity. In this section we compare our empirical results with the structural properties previously documented in the literature.

Degree Distributions. MO activity is known to follow a heavy–tailed distribution, with a small minority of expert users responsible for the majority of answers [3]. Our results reproduce this pattern: both in–degree and out–degree distributions exhibit heavy tails, with a few nodes reaching very high degree values, while most users appear with lower degrees. This confirms that the inequality of participation observed in the full network is also preserved in our filtered temporal subset.

Density and Clustering. Montoya et al. report that MathOverflow is sparse ($\rho \approx 0.014$) yet moderately clustered, with a global clustering coefficient around $C \approx 0.09$ [1]. Our subset shows nearly identical values ($\rho \approx 0.01$, $C = 0.09$), indicating that the temporal filtering and the activity threshold do not disrupt the topical modularity of the network. The persistence of non–trivial clustering suggests that collaborative reasoning still occurs within specialized mathematical sub communities, even in the restricted time window.

Reciprocity. MathOverflow interactions are predominantly unidirectional: users typically answer questions rather than engaging in extended bidirectional conversation [3]. While the literature does not report a specific value for reciprocity, our measured reciprocity of 0.39 aligns with this qualitative description. Most edges are non–mutual, reflecting the hierarchical expert–to–novice information flow that characterizes the platform.

Assortativity. MO exhibits negative assortativity: high–degree users tend to interact with low–degree users rather than with other high–degree peers [1]. Our analysis confirms this trend, with all assortativity measures (undirected, in–in, out–out and out–in) yielding negative coefficients. This reinforces the interpretation that MO is a expertise–stratified network, in which a core of highly active experts distributes knowledge to a large population of less active users.

Community Structure. Prior work characterizes MathOverflow as a thematically organized network, with moderate modularity: users cluster around mathematical subfields, yet extensive cross–topic interaction prevents strong fragmentation [2]. Our results reflect this behavior. The Louvain algorithm detects seven communities with a modularity of $Q = 0.2451$, consistent with the "moderate" structure described in the literature. The two dominant communities (549 and 509 nodes) alongside several smaller clusters mirror the coexistence of broad mathematical domains and specialized expert groups observed in earlier studies. Overall, the community structure we observe aligns closely with reported MO dynamics, supporting the validity of the filtered subset as a representative sample of the platform's organization.

Overall, the structural behavior of our filtered subset closely matches the properties described in the literature: heavy–tailed degree distributions, sparsity, moderate clustering, moderate modularity, small–world characteristics (as observed in the undirected projection), low reciprocity and consistently negative assortativity. The preservation of these core structural properties, despite temporal and activity-based filtering, confirms that the subset remains a faithful and representative sample of the underlying dynamics of the MathOverflow community.

6.2 Representativeness of the Subset

Our filtering procedure focused on two dimensions: temporal restriction (2010–2012) and user activity thresholds. The temporal window was chosen based on the observed surge in interaction volume during this period, which marks the most stable and active phase of MathOverflow's. The activity filter, retaining users with more than 25 interactions, effectively removes peripheral accounts while preserving the core contributor population that drives the majority of mathematical discussion.

The alignment between our measured structural properties and those reported for the full historical dataset supports the representativeness of the subset. Degree distributions, clustering, assortativity and community structure all match the qualitative and quantitative observations from prior work. This suggests that the chosen subset captures the behavior the platform, despite the pruning done.

6.3 Limitations and Future Work

Despite its representativeness, the subset-based approach has limitations. First, temporal filtering necessarily excludes long-range interactions and users whose activity falls outside the 2010–2012 window, potentially omitting evolving behavioral patterns such as community growth or shifting topical trends. Future work could incorporate sliding windows or longitudinal models to capture temporal evolution more fully.

Second, the activity threshold removes low-engagement users who may still play contextual or bridging roles. Alternative filtering strategies could preserve more peripheral structure while avoiding noise.

Finally, the absence of content-based features (question tags, topics, reputation scores) limits our ability to interpret communities semantically. This would allow deeper analysis of the thematic organization of the platform.

Overall, while the current subset enables a robust structural analysis, future extensions could incorporate temporal, semantic and behavioral dimensions to provide a more complete understanding of knowledge exchange in MathOverflow.

7 Conclusion

In this work, we analyzed the structural dynamics of the MathOverflow network (2010–2012). Our findings characterize the community as a robust, hierarchical Small-World network: short-average paths, medium-high clustering, and a strongly connected giant component. Despite the complexity of the full graph, the subset used still preserved these properties, indicating that the core patterns are stable even under filtering. The low reciprocity ($\sim$ 39%) and negative assortativity indicate a clear expert-to-novice knowledge transfer structure, rather than a flat discussion forum. While degree analysis highlighted the volume of contributions, the Betweenness Centrality analysis revealed a distinct set of "knowledge brokers" crucial for bridging the 7 communities identified. By combining temporal filtering with activity-based sampling, we also revealed clear behavioral patterns: user activity follows a pronounced weekly cycle and peaks during academic weekdays, suggesting an influence of researchers' professional schedules on community engagement. Future work should focus on semantic analysis, correlating the detected communities with actual tags (e.g., Algebra, Topology) to validate if these structural clusters correspond to mathematical subfields.

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